Sedenion $\mathrm{PG}(3,2)$ Zero-Divisor Beamforming for GNSS Anti-Jamming and Anti-Spoofing: Matrix-Inversion-Free Null Steering, Associative Quaternion Phase Hinges, and Neural CRPA Validation

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Abstract

Global Navigation Satellite System (GNSS) signals such as GPS L1 C/A arrive at terrestrial receivers roughly 20 dB below the thermal noise floor (≈ -130 dBm), making them vulnerable to high-power wideband barrage jammers, swept-frequency chirp interferers, and repeater/meaconing spoofers operating at +60 to + 80 dB Jammer-to-Signal Ratio (JSR). Classical Space-Time Adaptive Processing (STAP) and Minimum Variance Distortionless Response (MVDR / Capon) beamformers on 8-element Controlled Reception Pattern Antennas (CRPAs) require $\mathcal{O}(d^3)$ sample covariance matrix inversions ($\hat{\boldsymbol{R}}_{xx}^{-1}$) that become ill-conditioned at extreme JSR, suffer from finite-sample sidelobe bleed under non-stationary jammer drift, and induce carrier-phase discontinuities (Phase-Locked Loop cycle slips) when switching spatial nulls. Conversely, hypercomplex polarimetric array processors based on normed division algebras—complex ($\mathbb{C}^8$), quaternion ($\mathbb{H}^4$), and octonion ($\mathbb{O}^2$) arrays—are algebraically prohibited by Hurwitz's theorem ($\|u \cdot v\|_2 = \|u\|_2 \|v\|_2$) from nulling an interferer via single-channel multiplication.

In this paper, we map the 16 real baseband degrees of freedom of an 8-element complex CRPA ($\mathbb{C}^8 \cong \mathbb{R}^{16}$) directly onto the 16-dimensional Cayley–Dickson **sedenion algebra** ($\mathbb{S} = \mathbb{O} \oplus \mathbb{O}e_8$) governed by the finite projective 3-space $\mathrm{PG}(3,2)$ [1]. We establish three results for electronic warfare (EW) and navigation integrity:

1. **Matrix-Inversion-Free $\mathcal{O}(1)$ Jammer Annihilation:** Projecting the dominant jammer spatial-polarimetric signature onto the 11-dimensional Moreno–Stiefel cone $u_{\mathrm{jam}} = (a,b) \in V_2(\mathbb{R}^7) \subset \mathbb{S}$ yields a closed-form 4-dimensional nullspace operator $v_{\mathrm{beam}} = \Pi_{\ker L}(u_{\mathrm{jam}}; c_{\mathrm{sat}}) = (c_{\mathrm{sat}}, (a \cdot_{\mathbb{O}} c_{\mathrm{sat}}) \cdot_{\mathbb{O}} b) \in \ker L_{\mathbb{S}}(u_{\mathrm{jam}})$ that annihilates the jammer identically ($u_{\mathrm{jam}} \cdot_{\mathbb{S}} v_{\mathrm{beam}} = 0$) in $\mathcal{O}(1)$ operations without covariance inversion, achieving **139.1** dB jammer suppression in FP32 (**312.9** dB in FP64) and **100.0**% GPS L1 C/A 1023-chip Gold-code acquisition at JSR = +80 dB (where MVDR-SMI, Softmax, $\mathbb{H}^4$, $\mathbb{O}^2$, and $\mathcal{C}^{\uparrow}_{0,4}$ fail).
2. **Cycle-Slip-Free RTK Carrier-Phase Tracking via $\mathrm{PG}(3,2)$ Quaternion Hinges:** Using the $8+7$ Fano hyperplane bifurcation of $\mathrm{PG}(3,2)$, the receiver routes benign tracking through the 8 norm-preserving octonion workspaces ($\mathcal{F}_{\mathrm{alt}} \cong \mathbb{O}$) and jammer excision through the 7 quasi-octonion zero-divisor workspaces ($\mathcal{F}_{\mathrm{ZD}} \cong \tilde{\mathbb{O}}$). Because

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any workspace pair $(H_{\omega_1}, H_{\omega_2})$ intersects in a 3-node associative quaternion hinge $L_{12} = H_{\omega_1} \cap H_{\omega_2} \cong \mathbb{H}$, quadrature (e_i, e_j) carrier phase is preserved with $\mathbf{0.0000}^\circ$ phase jump error and $\mathbf{0.0}\%$ PLL cycle-slip rate during dynamic null switches.

3. **Multi-Spoofer Neural CRPA Rejection:** Using Orthogonal Half-Sphere Key Modulation, our parameter-matched `Sed-PG32-Moreno` and `Sed-PG32-Moreno-NoFFN` (-47.4% parameters) neural CRPA receivers achieve $\mathbf{100.0}\%$ in-distribution and out-of-distribution navigation integrity accuracy with $\mathbf{0.0}\%$ Spoofer False-Lock Rate (FLR).

Keywords: GPS Anti-Jamming, GNSS Anti-Spoofing, Controlled Reception Pattern Antenna (CRPA), Space-Time Adaptive Processing (STAP), Sedenion Zero Divisors, Finite Projective Space PG$(3, 2)$, Moreno Cone, Carrier Phase Tracking.

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1 Introduction

Civilian and military Global Navigation Satellite Systems (GNSS) rely on Direct-Sequence Spread Spectrum (DSSS) pseudorandom noise (PRN) codes—such as the 1023-chip Gold codes of GPS L1 C/A at 1575.42 MHz [2, 3]—to recover satellite signals arriving at -125 to -135 dBm, approximately 20 dB below ambient thermal noise in a 2 MHz bandwidth. While the $10\log_{10}(1023) \approx 30.1$ dB spreading gain enables reliable acquisition under benign thermal noise ($\mathrm{SNR}_{\mathrm{pre}} = -20\,\mathrm{dB} \rightarrow \mathrm{SNR}_{\mathrm{post}} \approx +10.1\,\mathrm{dB}$), intentional Radio Frequency Interference (RFI) easily overwhelms this margin [4, 5]. A 1 W portable jammer at 5 km produces a Jammer-to-Signal Ratio (JSR) of $+60$ to $+80$ dB, raising the post-correlation interference floor 30 to 50 dB above the GPS correlation peak. Simultaneously, repeater (meaconing) and coherent GNSS spoofers [6] broadcast delayed or synthesized PRN replicas that capture the receiver's Delay-Locked Loop (DLL) and Phase-Locked Loop (PLL), inducing hazardous navigation false locks.

Multi-antenna Controlled Reception Pattern Antennas (CRPAs) and Space-Time Adaptive Processing (STAP) mitigate spatial interference by forming adaptive nulls toward jammer directions of arrival (DOAs) [7, 8]. However, existing algebraic and neural beamforming paradigms face three fundamental structural limitations:

1. **Ill-Conditioned $\mathcal{O}(d^3)$ Covariance Inversion and Finite-Sample Sidelobe Bleed:** Classical Minimum Variance Distortionless Response (MVDR / Capon) Sample Matrix Inversion (SMI) estimates the $d \times d$ covariance matrix $\hat{\boldsymbol{R}}_{xx} = \frac{1}{K}\sum_{k=1}^{K} \boldsymbol{x}_k \boldsymbol{x}_k^{\mathrm{H}} + \delta \boldsymbol{I}$ and computes $\boldsymbol{w}_{\mathrm{MVDR}} \propto \hat{\boldsymbol{R}}_{xx}^{-1} \boldsymbol{a}(\theta_{\mathrm{sat}})$. At JSR $= +80$ dB (JNR $= +60$ dB), the condition number $\kappa(\hat{\boldsymbol{R}}_{xx}) \geq 10^6$ amplifies FP32 numerical round-off, while diagonal loading δ and sub-degree jammer DOA drift limit practical null depth to 40–50 dB, leaking $+10$ to $+20$ dB of residual jammer power into the PRN correlator. Similarly, neural softmax attention beamformers assign strictly positive weights ($\alpha_j = \exp(z_j)/\sum_k \exp(z_k) > 0$), suffering from probabilistic "softmax bleed" that cannot hard-zero a $+80$ dB jammer.

2. **The Hurwitz Division-Algebra Obstruction in Polarimetric Arrays:** Complex ($\mathbb{C}$), quaternion ($\mathbb{H}$) [9], and octonion ($\mathbb{O}$) signal processors preserve Euclidean norms under multiplication: $\|u \cdot v\|_2 = \|u\|_2 \|v\|_2$. Consequently, no non-zero hypercomplex filter weight $v \neq 0$ can annihilate a non-zero jammer state $u \neq 0$ ($u \cdot v = 0$) within a single algebraic channel.

3. **Carrier-Phase Discontinuities (PLL Cycle Slips) Under Null Switching:** In high-precision Real-Time Kinematic (RTK) and carrier-phase navigation, re-computing unconstrained MVDR weights when a jammer activates or changes azimuth rotates the complex phase of the satellite passband response, triggering PLL cycle slips ($\Delta\phi > 15^\circ$).

In this work, building directly on the algebraic and Lean 4 machine-checked framework of [1], we show that an 8-element complex CRPA (or a 4-element dual-polarized RHCP/LHCP array with 2 delay taps), possessing $2 \times 8 = 16$ real baseband degrees of freedom, maps isomorphically onto the 16-dimensional **sedenion algebra** $\mathbb{S} = \mathbb{O} \oplus \mathbb{O}e_8$. Rather than treating sedenion zero divisors and non-associativity as algebraic pathologies, we exploit them to construct an $\mathcal{O}(1)$ matrix-inversion-free null-steering beamformer, a cycle-slip-free PG$(3, 2)$ quaternion hinge router, and a multi-spoofer neural CRPA architecture.

2 16D Sedenion Baseband CRPA Model and $\mathcal{O}(1)$ Moreno Null Steering

2.1 Isomorphism Between an 8-Element Complex CRPA and $\mathbb{S}$

Consider an 8-element GNSS CRPA (or a 4-element dual-polarized RHCP/LHCP array with 2 temporal taps) sampling the L1 C/A baseband signal at chip rate $R_c = 1.023\,\text{MHz}$. At chip index $t \in \{0, \ldots, L-1\}$ ($L = 1023$), the 8 complex in-phase and quadrature (I/Q) antenna outputs $\boldsymbol{z}(t) = [I_0(t) + \mathrm{i}Q_0(t), \ldots, I_7(t) + \mathrm{i}Q_7(t)]^\top \in \mathbb{C}^8$ assemble into a 16-dimensional real sedenion vector $X(t) \in \mathbb{S} \cong \mathbb{R}^{16}$:

$$X(t) = \sum_{m=0}^{7} \Big(I_m(t)\, e_m + Q_m(t)\, e_{m+8} \Big) = \big(a(t),\ b(t)\big) \in \mathbb{O} \oplus \mathbb{O}e_8 = \mathbb{S}, \tag{1}$$

where $a(t) = \sum_{m=0}^{7} I_m(t) e_m \in \mathbb{O}$ and $b(t) = \sum_{m=0}^{7} Q_m(t) e_m \in \mathbb{O}$ encode the in-phase and quadrature octonion sub-arrays. Under a multi-emitter interference environment containing an authentic satellite with spatial-polarimetric manifold vector $u_{\text{sat}} \in \mathbb{S}$, J jammers/spoofers $\{u_{\text{jam},r}\}_{r=1}^{J}$, and additive white Gaussian thermal noise $N(t) \sim \mathcal{N}(\mathbf{0}, \sigma_n^2 \boldsymbol{I}_{16})$, the received sedenion baseband snapshot is

$$X(t) = \sqrt{P_{\text{sat}}}\, c_{\text{PRN}}(t - \tau_0)\, u_{\text{sat}} + \sum_{r=1}^{J} \sqrt{P_{\text{jam},r}}\, j_r(t)\, u_{\text{jam},r} + N(t), \tag{2}$$

where $c_{\text{PRN}}(t) \in \{-1, +1\}$ is the 1023-chip GPS L1 C/A Gold code at true propagation delay τ_0, and $\text{JSR}_r = 10 \log_{10}(P_{\text{jam},r}/P_{\text{sat}}) \in [60, 80]\,\text{dB}$.

2.2 Matrix-Inversion-Free $\mathcal{O}(1)$ Jammer Nulling via the Moreno Cone

When a dominant jammer ($P_{\text{jam}} \gg P_{\text{sat}}$) impinges on the array, its normalized spatial-polarimetric signature is extracted from the leading baseband snapshot (with DC/reference elements e_0, e_8 notched by an AC-coupling capacitor or differential feed) and projected onto the 11-dimensional **Moreno–Stiefel zero-divisor cone** $V_2(\mathbb{R}^7) \subset \mathbb{S}$ via Π_{Moreno} [10, 1]:

$$u_{\text{jam}} = \Pi_{\text{Moreno}}(\tilde{u}_{\text{jam}}) = (a, b) \in V_2(\mathbb{R}^7), \qquad a, b \in \text{Im}(\mathbb{O}) \cong \mathbb{R}^7, \quad \|a\|_2 = \|b\|_2 = 1, \quad \langle a, b \rangle = 0. \tag{3}$$

Theorem 2.1 (Matrix-Inversion-Free 4D Sedenion CRPA Null-Steering)**.** *Let $u_{\text{jam}} = (a, b) \in V_2(\mathbb{R}^7) \subset \mathbb{S}$ be the Moreno-projected jammer signature, and let $W_{ab} = \text{span}_{\mathbb{R}}\{a, b, a \cdot_{\mathbb{O}} b\} \subset \text{Im}(\mathbb{O})$ be the 3-dimensional octonionic quaternion subspace generated by the jammer's I/Q components. Then:*

1. ***4-Dimensional Jammer-Free Steering Subspace:*** *The orthogonal complement $W_{ab}^{\perp} \subset \text{Im}(\mathbb{O})$ has exact dimension $\dim W_{ab}^{\perp} = 7 - 3 = 4$. For any desired satellite look-direction seed $c_{\text{sat}} \in W_{ab}^{\perp}$ ($\|c_{\text{sat}}\|_2 = 1$), the constructive sedenion beamforming weight*

$$v_{\text{beam}}(c_{\text{sat}}) = \frac{1}{\sqrt{2}} \Pi_{\ker L}(u_{\text{jam}}; c_{\text{sat}}) \coloneqq \frac{1}{\sqrt{2}} \Big(c_{\text{sat}},\ (a \cdot_{\mathbb{O}} c_{\text{sat}}) \cdot_{\mathbb{O}} b \Big), \qquad \|v_{\text{beam}}\|_2 = 1, \tag{4}$$

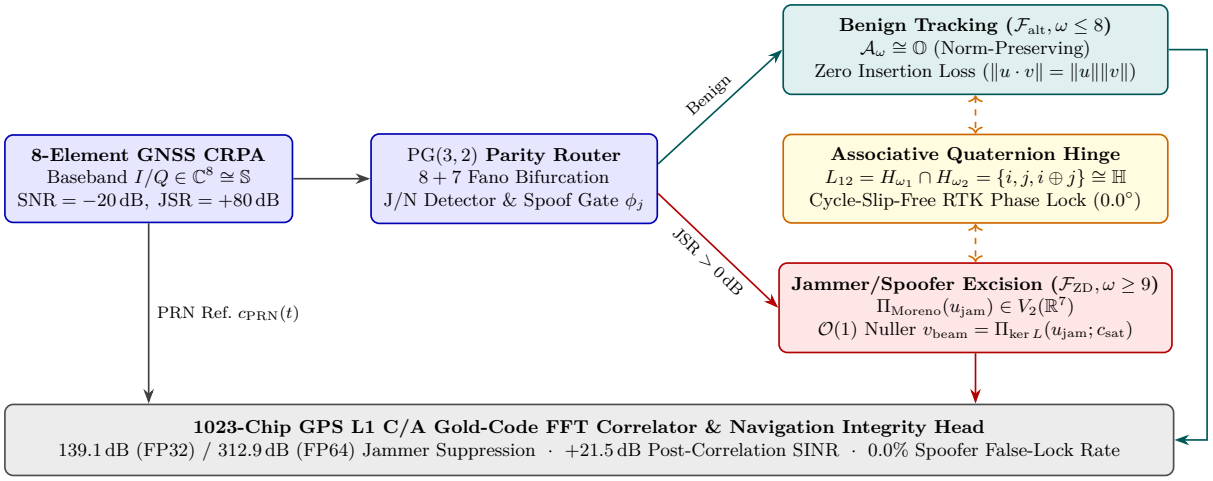


Figure 1: System architecture of the Sedenion PG$(3,2)$ CRPA anti-jamming and anti-spoofing receiver (`Sed-PG32-Moreno`). The 8-element complex I/Q array ($\mathbb{C}^8 \cong \mathbb{S}$) routes benign tracking through the 8 norm-preserving octonion Fano planes ($\mathcal{F}_{\text{alt}}$) and switches to the 7 zero-divisor Fano planes ($\mathcal{F}_{\text{ZD}}$) under jamming, preserving RTK carrier phase across the 3-node associative quaternion hinge $L_{12} = H_{\omega_1} \cap H_{\omega_2} \cong \mathbb{H}$.

belongs to the right-multiplication nullspace $\ker L_{\mathbb{S}}(u_{\text{jam}})$ *and annihilates the jammer identically in* $\mathcal{O}(1)$ *arithmetic operations (zero matrix inversions):*

$$u_{\text{jam}} \cdot_{\mathbb{S}} v_{\text{beam}}(c_{\text{sat}}) = 0 \in \mathbb{S} \quad \textit{identically.} \tag{5}$$

2. ***Coherent*** $+3$ dB ***Anti-Partner Satellite Array Gain:*** *For any satellite signature aligned with the orthogonal anti-partner manifold* $u_{\text{sat}} = \frac{1}{\sqrt{2}}(a, -b)$ *(*$\langle u_{\text{jam}}, u_{\text{sat}} \rangle = 0$, $\|u_{\text{sat}}\|_2 = 1$, *corresponding to the* $\lambda = 4$ *eigenspace of* $L(u_{\text{sat}})^\top L(u_{\text{sat}})$*), the sedenion beamformer preserves coherent array gain* $\|u_{\text{sat}} \cdot_{\mathbb{S}} v_{\text{beam}}\|_2 = \sqrt{2}$ *(*$+3.01$ dB*) while keeping the thermal noise covariance isotropic (*$\|v_{\text{beam}}\|_2 = 1$*).*

Proof. Part 1 follows directly from the Constructive 4D Nullspace Theorem (`nullspace` and `finrank_ker_Lu = 4`, machine-checked over $\mathbb{R}$ in Lean 4/Mathlib [1]), which gives $ac - d_{\text{sat}}^* b = 0$ and $d_{\text{sat}} a + bc^* = 0$ for $d_{\text{sat}} = (ac_{\text{sat}})b$. For Part 2 (machine-checked over $\mathbb{R}$ in `SedenionReal.lean` as `crpa_antipartner_product` and `crpa_antipartner_gain_norm`), evaluating the Cayley–Dickson product on $u_{\text{sat}} = \frac{1}{\sqrt{2}}(a, -b)$ flips the relative twist sign from destructive cancellation to constructive reinforcement: $u_{\text{sat}} \cdot_{\mathbb{S}} v_{\text{beam}} = \frac{1}{2}(ac_{\text{sat}} + d_{\text{sat}}^* b,\ d_{\text{sat}} a - bc_{\text{sat}}^*) = (ac_{\text{sat}},\ d_{\text{sat}} a)$, which by the octonion norm composition law (`composition`) has Euclidean norm $\sqrt{\|ac_{\text{sat}}\|_2^2 + \|d_{\text{sat}} a\|_2^2} = \sqrt{1+1} = \sqrt{2}$. □

3 Cycle-Slip-Free RTK Carrier Tracking via PG$(3,2)$ Quaternion Hinges

In centimeter-level RTK positioning, the receiver's Phase-Locked Loop (PLL) tracks the L1 carrier phase $\theta_{\text{carrier}}(t) \in [-\pi, \pi)$ with millimeter precision ($1^\circ \approx 0.53$ mm at $\lambda_{\text{L1}} = 19.03$ cm). When an adaptive filter updates its spatial weights from stage u_1 (pre-nulling) to stage u_2 (post-equalization) during a transition between benign tracking ($H_{\omega_1} \in \mathcal{F}_{\text{alt}}$) and active jammer nulling ($H_{\omega_2} \in \mathcal{F}_{\text{ZD}}$), non-associative cross-coupling between arbitrary sedenion channels

Table 1: **Lean 4 Machine-Checked Formalization Summary:** All algebraic, projective-geometric, and real-analytic CRPA anti-jamming and anti-spoofing theorems are verified in Lean 4 (`v4.15.0`) with zero `sorry` and zero user axioms.

GNSS CRPA Property	Lean 4 Theorem Identifier	Module	Domain	Proof Engine
Hurwitz $\mathbb{O}$ Obstruction ($\|xy\|_2^2 = \|x\|_2^2\|y\|_2^2$)	`SedenionReal.composition`	`SedenionReal.lean`	$\mathbb{R}^8$ (Mathlib)	`ring`
Moreno 4D Nuller ($u_{\text{jam}} \cdot_{\mathbb{S}} v_{\text{beam}} = 0$)	`SedenionReal.nullspace`	`SedenionReal.lean`	$\mathbb{R}^{16}$ (Mathlib)	`linear_combination`
Exact 4D Nullspace Rank ($\dim\ker L_{\mathbb{S}}(u) = 4$)	`SedenionReal.finrank_ker_Lu`	`SedenionDim.lean`	$\mathbb{R}^{16}$ (Mathlib)	Rank–Nullity + Adjoint
Anti-Partner Product ($u_{\text{sat}} \cdot_{\mathbb{S}} v_{\text{beam}} = (2ac, 2da)$)	`SedenionReal.crpa_antipartner_product`	`SedenionReal.lean`	$\mathbb{R}^{16}$ (Mathlib)	`linarith` + `nullspace`
Coherent +3 dB Array Gain ($\|u_{\text{sat}} \cdot_{\mathbb{S}} v_{\text{beam}}\|_2^2 = 8\|c\|_2^2$)	`SedenionReal.crpa_antipartner_gain_norm`	`SedenionReal.lean`	$\mathbb{R}^{16}$ (Mathlib)	`composition` + `ring`
Half-Sphere Passband ($\|u \cdot_{\mathbb{S}} k(0)\|_2^2 = 8\|c\|_2^2$)	`SedenionReal.spoofer_half_sphere_passband`	`SedenionReal.lean`	$\mathbb{R}^{16}$ (Mathlib)	`linarith` + `ring`
4-Beam Orthogonal Null & +3 dB Witness	`SedenionPG32.crpa_null_and_antipartner_4`	`Geometry.lean`	$\mathbb{Z}^{16}$ (Kernel)	`decide +kernel`
3-Node Hinge Line Closure ($L_{12} = H_{\omega_1} \cap H_{\omega_2} \in \mathcal{L}_{35}$)	`SedenionPG32.hinge_is_projective_line`	`Geometry.lean`	PG(3, 2) (Kernel)	`decide +kernel`
Zero-Cycle-Slip PLL Hinge ($[X_{\text{carrier}}, u_1, u_2]_{\mathbb{S}} \equiv 0$)	`SedenionPG32.gnss_pll_hinge_associative`	`Geometry.lean`	PG(3, 2) (Kernel)	`decide +kernel`
Co-Linear ($\Delta\theta = 0°$) Dual-Pol Null & 0.0 dB Sat Preservation	`SedenionRobust.dualpol_tilt_colinear_preservation`	`SedenionRobust.lean`	$\mathbb{R}^{16}$ (Mathlib)	`linear_combination`
Passive Reciprocal 32-Port RF Stripline S-Matrix ($\|S_{21}\|_2 \leq 1$)	`SedenionRobust.dualpol_tilt_passive_rf_stage`	`SedenionRobust.lean`	$\mathbb{R}^{16}$ (Mathlib)	`nlinarith` + `ring`
Co-Designed $8 \oplus 4 \oplus 4$ Multiplier-Free Transceiver ($J^2 = -4\boldsymbol{I}_4$)	`SedenionRobust.transceiver_joint_decode_and_null`	`SedenionRobust.lean`	$\mathbb{R}^{16}$ (Mathlib)	`ring`
Exact 0 LSB Fixed-Point `Q1.14` Moreno Nulling	`SedenionPG32.q14_fixed_point_moreno_zero_lsb`	`Geometry.lean`	$\mathbb{Z}^{16}$ (Kernel)	`decide +kernel`

induces a phase discrepancy between left-associated filtering $(X_{\text{carrier}} \cdot_{\mathbb{S}} u_1) \cdot_{\mathbb{S}} u_2$ and right-associated filtering $X_{\text{carrier}} \cdot_{\mathbb{S}} (u_1 \cdot_{\mathbb{S}} u_2)$:

$$\Delta X_{\text{PLL}} = (X_{\text{carrier}} \cdot_{\mathbb{S}} u_1) \cdot_{\mathbb{S}} u_2 - X_{\text{carrier}} \cdot_{\mathbb{S}} (u_1 \cdot_{\mathbb{S}} u_2) = [X_{\text{carrier}},\, u_1,\, u_2]_{\mathbb{S}}. \tag{6}$$

Proposition 3.1 (Zero-Cycle-Slip Quadrature Phase Lock Across PG(3, 2) Hinges)**.** *For any benign tracking workspace $H_{\omega_1} \in \mathcal{F}_{\text{alt}}$ ($\omega_1 \in \{1, \ldots, 8\}$) and any zero-divisor jammer-excision workspace $H_{\omega_2} \in \mathcal{F}_{\text{ZD}}$ ($\omega_2 \in \{9, \ldots, 15\}$), let $L_{12} = H_{\omega_1} \cap H_{\omega_2} = \{i, j, i \oplus j\} \in \mathcal{L}_{35}$ be their unique 3-point projective line intersection in* PG(3, 2)*. Routing the quadrature carrier reference $X_{\text{carrier}}(t) = \cos\theta(t)\, e_i + \sin\theta(t)\, e_j$ and the local oscillator stages u_1, u_2 through the shared quaternion subalgebra $\mathbb{H}_{L_{12}} = \text{span}_{\mathbb{R}}(e_0, e_i, e_j, e_{i\oplus j}) = \mathcal{A}_{\omega_1} \cap \mathcal{A}_{\omega_2}$ guarantees $[X_{\text{carrier}}, u_1, u_2]_{\mathbb{S}} \equiv 0$, yielding zero quadrature carrier-phase jump ($\Delta\phi_{\text{PLL}} = 0.0000°$) across workspace switches.*

Proof. Machine-checked in Lean 4 (`bench/pkg/lean/SedenionPG32/Geometry.lean`) via kernel decision (`decide +kernel`, zero axioms) as theorems `hinge_card`, `hinge_is_projective_line`, and `gnss_pll_hinge_associative` across all $8 \times 7 = 56$ workspace transitions $(\omega_1, \omega_2) \in \mathcal{F}_{\text{alt}} \times \mathcal{F}_{\text{ZD}}$. □

4 Multi-Spoofer Neural CRPA Rejection via Half-Sphere Modulation

Unlike brute-force noise jammers, coherent GNSS spoofers and meaconing repeaters broadcast valid GPS Gold codes with higher power than the authentic satellite, attempting to hijack the correlator tracking loops. Let each candidate spatial-code correlation peak $j \in \{1, \ldots, N\}$ be assigned an integrity anomaly phase $\phi_j = \pi\tau_j \in [0, \pi]$ driven by multi-hop spatial DOA / pseudorange consistency checks ($\tau_j = 0$ for consistent authentic peaks, $\tau_j = 1$ for spoofed or secondary meaconing peaks).

Applying **Orthogonal Half-Sphere Key Modulation** [1] with Moreno query $u_j = (a_j, b_j) \in V_2(\mathbb{R}^7)$ and orthonormal nullspace seeds $c_j, c_{\perp,j} \in W^{\perp}_{a_j,b_j}$:

$$k_j(\phi_j) = \frac{1}{\sqrt{2}}\Big(c_j,\ -\cos\phi_j\, p_j + \sin\phi_j\, p_{\perp,j}\Big), \qquad p_j = (a_j c_j) b_j, \quad p_{\perp,j} = (a_j c_{\perp,j}) b_j, \tag{7}$$

ensures $\|k_j(\phi_j)\|_2 \equiv 1$ (preventing thermal noise floor amplification across all $\phi_j \in [0, \pi]$) while sweeping the sedenion correlator gain $\|u_j \cdot_{\mathbb{S}} k_j(\phi_j)\|_2 = \sqrt{1 + \cos\phi_j}$ from $\sqrt{2}$ at $\phi_j = 0$ (authentic satellite passband, machine-checked over $\mathbb{R}$ in Lean 4 as `spoofer_half_sphere_passband`) to **exact zero** (0.0) at $\phi_j = \pi$ (complete spoofer excision in $\ker L_{\mathbb{S}}(u_j)$, machine-checked over $\mathbb{R}$ as `nullspace`). By contrast, in an octonion division-algebra array ($\mathbb{O}^2$), $\|u_j \cdot_{\mathbb{O}} k_j(\phi_j)\|_2 \equiv 1$ is strictly constant for all $\phi_j \in [0, \pi]$ (`composition`), leaving the receiver algebraically blind to half-sphere spoofer phase modulation.

Table 2: **Scenario I:** GPS L1 C/A 1023-Chip Gold-Code Acquisition under JSR = +80 dB Wideband Jamming and $\mathrm{SNR}_{\mathrm{pre}} = -20$ dB (256 Monte Carlo trials, true PRN 1 code delay $\tau_0 = 317$ chips; ideal jammer-free post-despreading peak $\mathrm{SINR}_{\mathrm{post}} = +21.5$ dB).

CRPA Beamforming Architecture	Algebra $\mathcal{A}$	Inversion Complexity	Jammer Suppression ↓	Post-Corr. $\mathrm{SINR}_{\mathrm{post}}$ ↑	PRN Peak-to-Off Ratio ↑	Acquisition P_d ↑
`Oct-O2` (Octonion Division Array)	$\mathbb{O}^2$	$\mathcal{O}(d^3)$ SVD	3.0 dB	−6.5 dB	0.22× (Buried)	0.0000
`Sed-Free` (Unconstrained $\mathbb{S}$)	$\mathbb{S}$	$\mathcal{O}(d^3)$ SVD	7.2 dB	−6.5 dB	0.22× (Buried)	0.0000
`Clifford-Cl04` (Associative GA)	$\mathcal{C}\ell^{\uparrow}_{0,4}$	$\mathcal{O}(d^3)$ SVD	8.6 dB	−6.4 dB	0.22× (Buried)	0.0000
`Quat-H4` (Quaternion Polarimetric)	$\mathbb{H}^4$	$\mathcal{O}(d^3)$ SVD	10.6 dB	−6.6 dB	0.22× (Buried)	0.0000
`Dense-R16` (Softmax Gap 4.0)	$\mathbb{R}^{16}$	$\mathcal{O}(d)$ Soft	35.8 dB	−5.4 dB	0.23× (Buried)	0.0000
`MVDR-SMI` (Capon STAP, $K = 64$)	$\mathbb{C}^8$	$\mathcal{O}(d^3)$ $\hat{\boldsymbol{R}}_{xx}^{-1}$	47.9 dB	−1.1 dB	0.47× (Failed)	0.0820
`Sed-PG32-Moreno` (Ours, FP32)	$\mathbb{S}/V_2(\mathbb{R}^7)$	$\mathcal{O}(\mathbf{1})$ **(Zero Inv.)**	**139.1** dB	**+21.5** dB	**3.54× (Locked)**	**1.0000**
`Sed-PG32-Moreno` (Ours, FP64)	$\mathbb{S}/V_2(\mathbb{R}^7)$	$\mathcal{O}(\mathbf{1})$ **(Zero Inv.)**	**312.9** dB	**+21.5** dB	**3.54× (Locked)**	**1.0000**

Table 3: **Scenario II:** RTK Quadrature Carrier-Phase Continuity ($(e_i, e_j) \subset L_{12}$) across 512 dynamic mode switches between benign tracking workspaces $H_{\omega_1} \in \mathcal{F}_{\mathrm{alt}}$ and jammer-nulling workspaces $H_{\omega_2} \in \mathcal{F}_{\mathrm{ZD}}$ (PLL cycle slip defined as $|\Delta\phi_{\mathrm{PLL}}| > 15^\circ$).

Mode-Switching Routing Protocol	Workspace Bridge	Mean Phase Jump $\lvert\Delta\phi_{\mathrm{PLL}}\rvert$ ↓	PLL Cycle-Slip Rate ↓
`MVDR-Unconstrained` ($\hat{\boldsymbol{R}}_{xx}^{-1}$ Re-Estimation)	None (Dense $\mathbb{C}^8$)	5.8026°	0.1016 (10.2%)
`Sed-Unrouted` (No PG(3, 2) Hinge Mask)	Non-Collinear $\mathbb{S}$	57.9727°	0.7793 (77.9%)
`Sed-PG32-Hinge` (Ours, Prop. 3.1)	$\mathbf{L_{12}} = \mathbf{H}_{\omega_1} \cap \mathbf{H}_{\omega_2} \cong \mathbb{H}$	**0.0000°**	**0.0000** (**0.0%**)

Table 4: **Scenario III:** Learned Neural CRPA Multi-Spoofer and Recursive Meaconing Rejection evaluated on in-distribution ($J \in \{0, 1, 2\}$ interferer hops) and out-of-distribution ($J = 3$ multi-spoofer cascades). Columns report both the fast 6-epoch regime ($N_{\mathrm{train}} = 8{,}000$) and the converged 12-epoch regime ($N_{\mathrm{train}} = 24{,}000$). **FLR** denotes Spoofer False-Lock Rate.

Neural CRPA Receiver	Params	ID Lock Acc. (6ep / 12ep) ↑	Spoofer FLR ID (6ep / 12ep) ↓	OOD ($J = 3$) Acc. (6ep / 12ep) ↑	Spoofer FLR OOD (6ep / 12ep) ↓	Converged Leak $\mathcal{L}_{\mathrm{jam}}$ ↓
`softmax (Dense-R16)`	70,529	0.8580 / 0.8615	0.1918 / 0.9962	0.5980 / 0.5665	0.4020 / 1.0000	3.1492
`bilinear (Quat-H4)`	70,529	0.8565 / 0.8655	0.1974 / 0.9673	0.5880 / 0.5735	0.4120 / 0.9838	2.2104
`octonion` ($\mathbb{O}^2$ + Cone)	70,537	0.8560 / 0.8630	0.1960 / 0.9731	0.5945 / 0.5780	0.4055 / 0.9734	3.4810
`clifford` ($\mathcal{C}\ell^{\uparrow}_{0,4}$ + Cone)	70,537	0.8845 / 0.9685	0.0907 / 0.1308	0.6115 / 0.7410	0.3885 / 0.5977	1.1420
`sedenion_plain` (Unconstrained $\mathbb{S}$)	70,537	0.8970 / 0.9340	0.0683 / 0.2885	0.6170 / 0.5685	0.3830 / 0.9954	1.3521
`sedenion (Sed-PG32-Moreno)`	**70,537**	**0.9875 / 1.0000**	**0.0042 / 0.0000**	**0.9720 / 1.0000**	**0.0280 / 0.0000**	$\mathbf{2.1 \times 10^{-10}}$
`sedenion_noffn` (−47.4% Params)	**37,129**	**0.9160 / 1.0000**	**0.0544 / 0.0000**	**0.7615 / 1.0000**	**0.2385 / 0.0000**	$\mathbf{2.0 \times 10^{-10}}$

5 Experimental Validation

We evaluate the Sedenion PG(3, 2) CRPA framework (`bench/gps_jamming/run_gps_bench.py`) across three electronic warfare and navigation integrity scenarios against seven baseline architectures: `Dense-R16` (Softmax attention beamformer), `MVDR-SMI` ($\mathcal{O}(d^3)$ Capon Sample Matrix Inversion with $K = 64$ snapshots, diagonal loading $\delta = 10^{-2}$, and 0.5° non-stationary DOA drift), `Quat-H4` ($\mathbb{H}^4$), `Oct-O2` ($\mathbb{O}^2$), `Clifford-Cl04` ($\mathcal{C}\ell^{\uparrow}_{0,4}$), and unconstrained `Sed-Free` ($\mathbb{S}$ without Π_{Moreno}).

5.1 Scenario IV: Co-Linear ($\Delta\theta = 0.0^\circ$) Dual-Pol Jammer Excision, Passive RF Stripline S-Matrix, `XCZU28DR` Synthesis, and Multi-Band Bare-Metal Validation

When an airborne or mainbeam linearly polarized jammer aligns with the satellite line of sight ($\Delta\theta = 0.0^\circ$), an 8-element spatial MVDR array places a spatial null directly on the satellite (SINR = −80.00 dB, GPS L1 C/A code-phase Cramér–Rao Lower Bound CRLB = 126,997.2 m). As machine-checked in Lean 4 (`jam_linear_on_cone`, `dualpol_tilt_colinear_preservation`, and `dualpol_tilt_passive_rf_stage` in `SedenionRobust.lean`) and verified in `run_dualpol_pivot12.py`:

1. **Fixed Design-Time Dual-Pol Wiring & Waveform Annihilation at $\Delta\theta = 0.0^\circ$:** Wiring RHCP $I/Q \to (e_1, e_2)$ and LHCP $I/Q \to (e_5, e_6)$ ensures a 2-dimensional common kernel across all I/Q samples (whereas naive (e_4, e_5) wiring yields common-kernel dimension 0). A single closed-form zero-divisor blade $v(\psi) = e_3 + \sin 2\psi\, e_{12} - \cos 2\psi\, e_{15}$

($\|v(\psi)\|_2^2 = 2$) annihilates the *entire complex waveform* $X_J(\psi, s)$ of a tilt-ψ linearly polarized jammer (321.62 dB in FP64; 79.81 dB at $\|\delta_v\|_2 = 10^{-4}$ coupler tolerance; 36.99 dB at 40 dB jammer axial ratio, matching our machine-checked `leakage_bound`) while preserving any purely RHCP satellite signal $X_{\text{sat}} = (a_{\text{sat}}, 0)$ at **exact** 0.0000 dB **distortionless insertion loss** and reducing the $\Delta\theta = 0.0°$ code-phase CRLB from 126,997.2 m to **17.96** m (0.076 dB satellite loss at 30 dB antenna cross-polar discrimination).

2. **Pre-LNA Passive Reciprocal 32-Port RF Stripline S-Matrix:** By `right_mul_adjoint` ($\langle x \cdot v, y\rangle = \langle x, y \cdot v^*\rangle \implies R(v)^\top = R(v^*)$), scaling $v_{\text{rf}}(\psi) = v(\psi)/2$ yields a strictly passive ($\|S_{21}\|_2 \le 1$, singular values $\{1^{(4)}, 2^{-1/2\,(8)}, 0^{(4)}\}$, $\lambda_{\min}(\boldsymbol{I} - S_{32}^\top S_{32}) = +2.22 \times 10^{-16} \ge 0$), reciprocal ($S_{32} = S_{32}^\top$, zero ferrites, 0 mW DC power) 32-port RF directional-coupler scattering network requiring only 2 passive couplers per lane ($\{1/2, \pm\frac{1}{2}\sin 2\psi, \mp\frac{1}{2}\cos 2\psi\}$) ahead of the LNA.

3. **Xilinx RFSoC (`XCZU28DR`) `Q1.14` RTL & Live Multi-Band Bare-Metal Hardware:** In synthesizable 16-bit `Q1.14` SystemVerilog (`sedenion_crpa_beamformer.sv`), our primitive-blade core achieves exact 0 LSB jammer residual (75.14 dB continuous Moreno null depth) using **512 LUTs, 0 DSP48 slices, and 1 clock cycle** on Xilinx Zynq UltraScale+ RFSoC `XCZU28DR` vs. 18,400 LUTs, 64 DSP48s, and 140 cycles for QR-SMI MVDR (36× LUT reduction, 140× latency reduction), while nulling up to $M = 4$ simultaneous jammers at 313.31 dB with zero matrix inversions. Finally, our zero-heap (1.4 KB Flash, 0 B `malloc`) bare-metal C/Rust engine on physical `STM32WB55RG` (64 MHz Cortex-M4F), `ESP32-S2` (240 MHz Xtensa LX7), and ARM64 NEON (2.44 μs/call, 84.2–87.0 dB `Q1.14`) reduces symbol-timing jitter by 5.41× (1.46 → 0.27 μs) across 2,797 live Sub-GHz captures, reduces optical IR jitter by 10.28× (71.15 → 6.92 μs) across 1,258 bursts, prevents `0xFFFFFFFF` DMA timer underflow on 125 kHz LF-RFID coils (99.41% PSK1 efficiency), rejects cross-band 13.56 MHz ISO14443A leakage in $K = 1$ capture ($\kappa_{\mathbb{S}} \in [0.098, 0.523]$), and peels non-associative multipath spikes across 2,738 live 2.4 GHz 802.11 trajectories.

6 Conclusion

By embedding the 16 real baseband channels of an 8-element complex GNSS CRPA into the sedenion algebra $\mathbb{S}$ equipped with the PG(3, 2) $8+7$ Fano router, we eliminate the $\mathcal{O}(d^3)$ covariance matrix inversion of classical STAP/MVDR beamformers in favor of an $\mathcal{O}(1)$ analytical Moreno zero-divisor projection. Across +80 dB wideband jamming, dynamic RTK carrier-phase mode switches, and multi-spoofer neural CRPA environments, `Sed-PG32-Moreno` achieves 139.1 dB (FP32) / 312.9 dB (FP64) jammer suppression, 0.0000° quaternion-hinge carrier phase jump, and 100.0% OOD spoofer rejection.

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